



FIRST SEMESTER

Marketing Analytics - COURSE HANDOUT

In addition to Part I (General Handout for all courses appended to the Time Table), this portion gives further specific details regarding the course.

Course No.: MPBA G524

Course Title: MARKETING ANALYTICS

Course Instructors: Prof. Achint Nigam (achint.nigam@pilani.bits-pilani.ac.in)

Course Description (Scope and Objective): This course equips students with the analytical frameworks, methods, and tools needed to make evidence-based marketing decisions across the customer lifecycle. It introduces the role of marketing analytics in supporting strategic and operational decisions, distinguishing business intelligence, analytics, and data science, and communicating insights through dashboards and data-driven storytelling.

Students will apply analytical techniques to core marketing problems including customer segmentation and positioning, product and pricing analytics, market mix modeling, customer journey analysis, customer experience, retention and lifetime value, digital marketing measurement, and attribution. The course covers methods such as cluster analysis, multidimensional scaling, conjoint analysis, regression, price elasticity modeling, forecasting and diffusion models, principal component analysis, logistic regression, A/B testing, ANOVA, Bayesian networks, and optimization.

The course also examines the growing role of artificial intelligence, machine learning, text and image analytics, recommender systems, personalization, and responsible generative AI in marketing. Emphasis is placed on practical interpretation and managerial application of results using tools such as Microsoft Excel, Power BI, DAX, and Python/R. By the end of the course, students will be able to translate marketing data into actionable insights, evaluate marketing performance, and develop data-informed recommendations for products, customers, channels, and brands.

Text Book:

- I. Marketing Analytics (2026), 2nd ed. by Seema Gupta, Avadhoot Jathar. Wiley (recommended purchase). https://www.amazon.in/Marketing-Analytics-2ed-Seema-Gupta/dp/9373320483/ref=tmm_pap_swatch_0

Reference Book:

- I. Marketing Analytics: Based on First Principles (2022). Authors: by Robert W. Palmatier, J. Andrew Petersen, Frank Germann. Bloomsbury Academic India.
- II. Python for Marketing Research and Analytics. First Edition. Springer. Jason S. Schwarz, Chris Chapman, and Elea McDonnell Feit

Session Plan

S.No.	Topic	Chapter Number	Learning Objectives
1.	Introduction to Marketing Analytics	1.	• Understand which marketing decisions are supported by analytics. • Learn what different analyses are important considering product categories. • Differentiate between business intelligence, analytics, and data science. • Learn about exploratory data analysis, predictive analytics, and prescriptive analytics. • Interpret integrated analytics using dashboards to support data-driven storytelling and decision-making. • Explore career opportunities and



			evolving roles in marketing analytics across industries and organizations.
2.	Segmentation	2.	- Understand the types and benefits of customer analytics. - Learn how data is collected and analyzed in segmentation. - Explore the process, applications, and examples of cluster analysis. - Perform calculations using clustering algorithms such as k-means. - Interpret and evaluate clusters effectively. - Understand hybrid
3.	Positioning	3.	- Define product positioning and understand its importance. - Construct various types of perceptual maps. - Recognize the importance of white spaces and learn how to leverage them. - Identify different types of product differentiation. - Understand umbrella branding and its strategic value. - Comprehend the multidimensional analysis technique.
4.	Product Analytics	4.	- Understand choice models. - Learn conjoint analysis—its principles, variables, steps, benefits, and supporting software. - Apply value-based pricing using customer-perceived value and market comparison. - Comprehend market forecasting and the S-curve. - Gain insights into the Bass diffusion model and adopter categories.
5.	Pricing	5.	- List and explain the goals of pricing. - Understand pricing strategies such as bundling, skimming, and revenue management. - Explain the relevance of pricing strategies to products and service categories. - Describe the concept of price promotions. - List the various types of price promotions and discounts. - Identify ways to evaluate which price promotions are effective. - Understand price elasticity modeling and how to use regression analysis for it.
6.	Marketing Mix	6.	- Understand the marketing mix. - Define market mix modeling (MMM), identify variables in MMM, and recognize data-related challenges. - Comprehend statistical modeling through regression analysis. - Understand optimization techniques used in market mix modeling. - Transform wide marketing data for dynamic Power BI visualizations.
7.	Customer Journey	7.	- Understand the customer journey and customer journey mapping, and explain the steps involved. - Identify touchpoints and explain their importance with examples. - Explain the role of analytics in customer relationship management, supported by examples. - Comprehend the customer journey for a consumer packaged goods (CPG) brand, including the process of data gathering, analysis, and interpretation. - Understand Principal Component Analysis and dimensionality reduction in survey data.
8.	Nurturing Customers	8.	- Define customer experience analytics and understand its goals. - Understand the different metrics used to track customer experience, such as net promoter score, customer effort score, churn rate, and upsell and cross-sell rates. - Understand logistic regression analysis, including how to use diagnostic tests and apply logistic regression as a classification technique. - Apply



			service metrics and sentiment analysis to enhance customer satisfaction and recommendation decisions.
9.	Customer Analytics	9.	• Understand buyer personas and how to build them. • Recognize the importance of calculating customer lifetime value and understand the process. • Learn strategies to boost customer lifetime value and identify profitable customers. • Understand the importance of customer retention and how to address customer churn. • Apply retention strategies using personalization and loyalty programs. • Apply optimizing product assortment planning. • Use DAX expressions with Power BI to visualize churn and customer lifetime value.
10.	Digital Analytics: Metrics and Measurement	10.	• Understand the metrics used in digital marketing and web analytics. • Explain the attribution challenge in marketing and understand how Shapley regression analysis can help address it. • Understand common visualizations used with these metrics and tools. • Learn how to use A/B testing and analysis of variance (ANOVA) for field experiments, which are common in digital marketing. • Explore the technique of Bayesian networks to understand probability-based relationships among various metrics.
11.	Artificial Intelligence and Machine Learning	11.	• Understand the tenets of deep learning (DL) and the components of an artificial intelligence (AI) system • Identify and comprehend marketing use cases of artificial intelligence and machine learning (ML) algorithms in direct marketing, personalization, and recommender systems. • Grasp the concepts involved in using text and image data for marketing applications. • Explore implementation details of two key use cases—targeting (direct marketing) and recommendations—and understand how Python libraries are used for these purposes. • Recognize the challenges associated with AI. • Understand generative AI concepts, tools, and techniques, and apply them responsibly across marketing, customer service, and domain-specific content generation tasks.
12.	Data Visualization	12.	• Understand the importance of visualization. • Create key visualization charts using univariate and multivariate data, as well as descriptive visualizations. • Learn essential visualization charts alongside selected analytics techniques. • Learn to craft brand narratives using Power BI by integrating metrics, associations, and product attributes.



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4. Evaluation Scheme: Evaluation components assess a range of learning outcomes: basic comprehension of the topics taught, application of different assignments in the lab skills learned, and ability to solve new data problems. The proposed evaluation scheme is given in the table below. It may be modified based on the prevailing conditions when the course begins.

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Quiz-1 (September)	30 minutes	10%	TBD	TBD
Quiz-2 (November)	30 minutes	10%	TBD	TBD
Group Project*	--	20%	TBD	TBD
Mid Term	1.5 hours	25%	TBD	TBD
Comprehensive	3 hours	35%	TBD	TBD
Class participation	--	--	--	--

*Group project: **Proposal submission: TBD. Mid review: TBD, Final submission: TBD.**

5. Chamber Consultation Hour: TBD

6. Notices: Notices will be sent to your official Email.

7. Make-up Policy: No make-ups will be granted for Projects, Labs and quizzes. For other components, proper medical certification from the institute's medical center will be essential. Makeups if any will be on IC discretion

8. Note:

- Carrying a laptop is mandatory. You will lose class participation marks for not bringing laptops to class.

ii. Attending class is optional but not being on time isn't.

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